RAJALAKSHMI ENGINEERING COLLEGE

Reg. No: 2116001203

Continuous Assessment-II	
Information Technology	
Course code & Name: AI23231 – Principles of Artificial Intelligence	
Semester: V	Year: III
Date: 27.10.2025	QP CODE: 101103

[Regulations: 2023]

Time: 02.00 hours Max. Marks: 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	What is the role of heuristic functions in the Minimax search process?	[CO3]	[k1]
2	Define iterative deepening.	[CO3]	[k1]
3	Define certainty factors and explain their relevance in expert systems.	[CO4]	[k1]
4	What are the advantages of using fuzzy logic in AI-based decision making?	[CO4]	[k2]
5	Define logic programming and give one real-time AI example.	[CO4]	[k1]
6	Compare forward reasoning and backward reasoning with respect to inference direction.	[CO4]	[k2]
7	Define unsupervised learning and state one example of its application.	[CO5]	[k1]
8	What is meant by reinforcement signal in reinforcement learning?	[CO5]	[k2]
9	List any two examples of classification algorithms used in machine learning.	[CO5]	[k2]
10	State one key difference between regression and clustering problems.	[CO5]	[k2]

Part B: 5x11=55 Marks

11	a.	Explain the working principle of the Alpha–Beta pruning algorithm using a suitable game tree example. Highlight how pruning affects computational complexity with a diagram.	[CO3]	[k 3]
		[OR]		
		Describe the Iterative Deepening Search strategy. Explain its advantages and limitations with a flow diagram showing node expansion at each depth.	[CO3]	[k 3]

12	a.	Explain the concept of Certainty Factors in rule-based systems. Discuss how they manage uncertainty in expert systems.	[CO4]	[k2]
		[OR]		
	b.	Describe the structure and operation of a Fuzzy Inference System (FIS) with a neat block diagram. Discuss its application in real-world AI decision processes.	[CO4]	[k2]
13	a.	Discuss rule-based and logic-based knowledge representation techniques. Explain how they are implemented in Prolog.	[CO4]	[k 2]
		[OR]		
	b.	Explain the process of knowledge inference using forward chaining and backward chaining.	[CO4]	[k2]
14	a.	Describe the types of machine learning paradigms. Compare them using a conceptual Venn diagram.	[CO5]	[k3]
		[OR]		
	b.	Illustrate the working principle of a reinforcement learning agent with the agent–environment feedback loop. Discuss an example from robotics.	[CO5]	[k 3]
15	a.	Explain the difference between classification and regression models. Use a graphical illustration to show how decision boundaries are formed.	[CO5]	[k2]
		[OR]		
	b.	Describe the architecture of a simple feed-forward neural network and explain its training process with a neural diagram showing weights and layers.	[CO5]	[k2]

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Reg. No.	2	1	1	6	2	4	1	5	0	1	0	8	0

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Continuous Assessment Test- II [CAT - II]

Year : I
Semester : II
Branch : AI&ML, AI&DS
Sub. Code : AI23231
Subject Name : Principles of Artificial Intelligence
QP Code : 011157

[Regulations 2023]

Date:10.06.2025 **Time: 120 Minutes** **Marks: 75**

Answer ALL Questions

Part A [10 x 2 = 20 Marks]

1	Define alpha-beta pruning in the context of game theory.	[A1]	[CO3]
2	What is meant by perfect and imperfect information in games?	[A2]	[CO3]
3	Name two quantifiers used in FOL. Give one example for each.	[A2]	[CO4]
4	State Modus Ponens with an example.	[A2]	[CO4]
5	Define rule-based knowledge representation.	[A1]	[CO4]
6	Explain the significance of forward and backward reasoning in AI.	[A2]	[CO4]
7	What is Baye's Theorem.	[A1]	[CO5]
8	Differentiate between chance node and decision node in a decision network.	[A2]	[CO5]
9	What is fuzzy logic? Mention one real world application.	[A2]	[CO5]
10	What is the Dempster-Shafer Theory and how does it differ from traditional probability theory?	[A1]	[CO5]

Part B [5 x 11 = 55 Marks]

11.	a.	Explore how game theory optimizes AI decision-making processes with examples, detailing the computational advantages and potential limitations.	[B1]	[CO3]
		[OR]		
	b.	Describe the Minimax algorithm used in adversarial search. Explain how it works with the help of a game tree. Also, Mention its Limitations.	[B2]	[CO3]

12.	a.	Explain Forward chaining and backward chaining in the context of rule-based system. Discuss their working mechanism and provide the example for each	[B1]	[CO4]
		[OR]		
	b.	Assess the real-world implementation of rule-based systems in commercial AI applications, analyzing their performance and areas for improvement.	[B2]	[CO4]
13.	a.	Explain propositional logic as knowledge representation techniques. Discuss its syntax, semantics and inference mechanism. Illustrate with example.	[B1]	[CO4]
		[OR]		
	b.	Define First order logic. How it differs from propositional logic? Explain the syntax and Semantics of FOL with example.	[B2]	[CO4]
14.	a.	Evaluate the incorporation of Dempster-Shafer Theory within medical diagnostic expert systems, focusing on its contributions to handling uncertainty.	[B1]	[CO5]
		[OR]		
	b.	Discuss how Bayesian Networks are applied in AI for statistical reasoning, providing examples of their effectiveness in real-world decision-making scenarios.	[B2]	[CO5]
15.	a.	Critically analyze how probabilistic reasoning is enhanced by the application of Certainty Factors, especially in complex diagnostic systems.	[B1]	[CO5]
		[OR]		
	b.	Explain decision network and Markov decision process as decision making tools for AI. Discuss their components, working principles and differences.	[B2]	[CO5]